

$$f^T f =$$

$$L \quad L^T$$

$$A = LL^T$$

$$L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix} \quad \begin{bmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{bmatrix}$$

$$L^T = \begin{bmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{bmatrix}$$

$$\Rightarrow \begin{cases} l_{11}^2 = 9 \\ l_{11} = 3 \end{cases} \quad \begin{cases} l_{11} l_{21} = 15 \\ = 3 \cdot l_{21} = 15 \\ l_{21} = 5 \end{cases}$$

$$LL^T = \begin{bmatrix} l_{11}^2 & l_{11} l_{21} \\ l_{21} l_{11} & l_{21}^2 + l_{22}^2 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 15 & 30 \end{bmatrix}$$

$$l_{21}^2 + l_{22}^2 = 30; \quad 25 + l_{22}^2 = 30$$

$$l_{22} = \sqrt{5} \text{ or } 5$$

$$\therefore L = \begin{bmatrix} 3 & 0 \\ 5 & 5 \end{bmatrix}; \quad L^T = \begin{bmatrix} 3 & 5 \\ 0 & 5 \end{bmatrix}$$